A project report on

BLOCKCHAIN-SECURED ACADEMIC CERTIFICATE VAULT

Submitted in partial fulfillment for the award of the degree of

INTEGRATED MTECH

by

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DECLARATION

I here by declare that the thesis entitled “BLOCKHAIN-SECURED ACADEMIC CERTIFICATE VAULT ” submitted by T, DHEDEEPYA, M.NIHARIKA, M .VASUNDHARA, for the award of the degree of Integarted m.tech VIT is a record of bonafide work carried out by me under the supervision of Dr.SHAIK KAREEMULLA.

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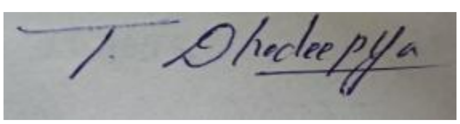

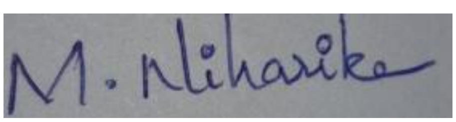

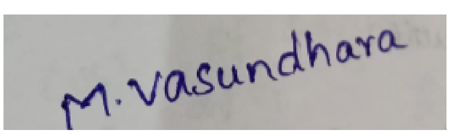

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ABSTRACT

Academic certificate fraud and ineffective verification procedures are still problems for employers and educational institutions worldwide in the current digital era. The Blockchain-Secured Academic Certificate Vault, a safe, decentralized platform for storing and confirming student academic records, is what this project suggests. By utilizing blockchain technology, the system eliminates the possibility of tampering or forgery and guarantees the authenticity and immutability of academic transcripts and certificates.

Cloud-based certificates are connected to blockchain-based digital fingerprints, or hashes, for validation. Employers and academic institutions can quickly confirm the legitimacy of any academic record using a cloud-hosted API rather than depending on laborious validation procedures. Students can manage who can view their credentials and for how long thanks to the system's time-limited, shareable access links, which protect privacy.

The platform offers a safe, effective, and user-friendly solution for managing and verifying academic records by integrating cloud computing, decentralized ledgers, and API-based access control. This project has a great chance of being implemented in the real world, particularly in fields where the authenticity of credentials is crucial.

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LIST OF ACRONYMS

Acronym	Abbreviation
DApp	Decentralized Application
IPFS	InterPlanetary File System
JSON	JavaScript Object Notation
NFT	Non-Fungible Token
SHA-256	Secure Hash Algorithm (256-bit)
EVM	Ethereum Virtual Machine
UI	User Interface
API	Application Programming Interface
Txn	Blockchain Transaction
Smart Contract	Self-executing code on blockchain
Wallet	Digital account for blockchain interactions

CHAPTER 1

INTRODUCTION

1.1 BACKGROUND AND MOTIVATION

Academic certificate authenticity has been one of the major concerns in this digital era. More and more cases of forging degrees, fraudulent certificates, and data tampering have undermined the legitimacy of institutions of learning. Conventional verification systems rely on manual checks, third-party intermediaries, and paper-based processes, which are time-consuming and prone to manipulate.

It provides an immutable, transparent, and decentralized solution to this problem. The storing of academic credentials on blockchain networks would enable institutions to issue tamper-proof certificates, and employers could verify them instantly with no intermediaries. This project, "**Academic Certificate Vault Secured by Blockchain**," aims to develop a cloud-hosted decentralized platform that securely stores and verifies academic certificates using blockchain and IPFS integration.

1.2 PROBLEM STATEMENT

Following are some of the challenges that lie in the current verification processes at academic institutions:

- With the advancement in editing tools, certificates can be forged easily.
- Universities undertake very slow and error-prone manual verification.
- Centralized databases are vulnerable to hacking or unauthorized access.
- Lack of a uniform standard of verification across different institutions.

Therefore, there is an urgent need for a tamper-proof, decentralized, and accessible system for the issuance and verification of academic certificates.

1.3 RESEARCH OBJECTIVES

The major objectives of the project are:

- To design a blockchain-based storage system that can issue and verify academic certificates.
- Integrate IPFS to store certificate files in a decentralized storage.
- In order to implement smart contracts for the purpose of issuance and verification in a transparent manner.
- To develop a cloud-hosted, web-based interface for the student, admin, and verifier roles.

Assess the efficiency of the system, speed of transactions, and ability to scale.

1.4 SCOPE OF THE PROJECT

This project focuses on the design of a decentralized certificate management system to be adopted by educational institutions. It includes:

- Blockchain-based verification of certificates.
- Integration with IPFS for secure file storage.
- Issuance and ownership tracking by smart contracts.Cloud deployment for universal access.

Future extensions could include integration with digital identity systems and multi-chain interoperability.

1.5 SIGNIFICANCE OF BLOCKCHAIN IN ACADEMIC CERTIFICATION

Blockchain ensures:

- **Immutability:** Once a certificate is stored, it cannot be altered or deleted.
- **Transparency:** Every transaction can be publicly verified.

- **Decentralized:** No controlling entity in the system minimizes the chances of **manipulation** .
- **Security** - Certificates are cryptographically signed and stored.

This creates digital trust between universities, students, and employers around the world.

1.6 CHALLENGES IN CERTIFICATE ISSUANCE AND VERIFICATION

Despite the increasing digital transformation in education systems, there are still a lot of problems with the management of academic certificates. First of all, the ease with which documents can be forged-for instance, fake degrees and mark sheets are made on some basic graphic tools and then sold for a price online-dictates that institutions rely on traditional, centralized databases or even manual archives, which are susceptible to data corruption, unauthorized access, and insider tampering. Verification generally requires third-party agencies or direct communications between institutions, thus making the process sluggish, bureaucratic, and costly on both ends-for graduates and employers alike.

Another important issue at stake is that of the interoperability of data. Various institutions have different systems, and these are usually not compatible; hence, storage and retrieval are fragmented. Furthermore, academic certificates in most developing countries are paper-based, leaving them prone to risks such as loss, theft, and physical damages. In cases where verifiers want to confirm authenticity, they have no way other than approaching the university that issued it. Employers have to wait for several days or even weeks to validate a single certificate.

Finally, there is trust and privacy. Many students are reluctant to publish their credentials online because they are afraid of data misuse, while the university cannot ensure verification transparency. These gaps call for a decentralized, tamper-proof, verifiable system—something that only blockchain technology can provide.

1.7 BLOCKCHAIN ADVANCES FOR SECURE DATA MANAGEMENT

Over the past decade, blockchain has grown from being merely a cryptocurrency ledger to a core technology capable of reshaping the very fabric of trust in digital ecosystems. The decentralized and distributed nature of blockchain provides the guarantee that data, once recorded, cannot be altered without consensus; hence, manipulation is eliminated. In education, blockchain has just begun to gain significant interest for verification of credentials, storage of transcripts, and management of digital identities**.**

Already, pioneering educational organizations like MIT, Harvard, and the University of Nicosia have tested the concept of blockchain-based diplomas, whereby secure storage and easy verification can be done by merely scanning a digital hash. In this way, this new paradigm not only fights document fraud but also enables global interoperability whereby any employer or institution can verify credentials without having to contact the university directly that has issued them. Blockchain further increases the level of automation through its smart contract capability: the logic can be embedded in contracts to enable automatic issuance upon course completion and instant verification upon request, or revocation if that be the case, without any human intervention. Coupled with cloud deployment, such systems become scalable, accessible, and cost-effective for mass adoption across institutions worldwide**.**

1.8 JUSTIFICATION FOR CHOOSING BLOCKCHAIN AND IPFS

Blockchain was chosen as a core technology for this project because it will enable immutability, security, and transparency, which are crucial in the safety of academic data. When the issuance of a certificate takes place, its unique hash value gets stored on the blockchain, so that even though the original document has modifications, the authenticity of the document can be assessed through hash comparison. Each transaction involved is time-stamped and gets recorded on every node in the network, making it impossible for any entity to manipulate or delete data.

However, blockchain is not designed to store large files efficiently. The project hence involves a combination with IPFS-a decentralized file storage network that distributes files across a pool of nodes. Original certificate PDFs are uploaded to IPFS to generate a unique content identifier (CID). The same is stored on the blockchain via a smart contract linking the certificate metadata to an immutable proof of existence.

This proposed approach allows for highly efficient storage, verifiability, and global accessibility. IPFS ensures that the document remains accessible even in the event of one node going offline, while blockchain provides the verifiable proof of authenticity of the document. Together they form a secure, scalable ecosystem of academic document management that outperforms centralized solutions in reliability, transparency, and protection of privacy.

1.9 PROJECT INNOVATIONS

This project introduces several innovative aspects, combining blockchain's transparency with the scalability of cloud-based systems:

- **Hybrid Blockchain-IPFS Architecture**: Incorporates blockchain's immutability with IPFS decentralized storage in handling certificate files effectively.
- **Smart Contract Automation:** Issuance, verification, and revocation of certificates are automated through smart contract functionality in Solidity.
- **Role-Based Access Control:** Students, universities, and verifiers all have different permissions in order to maintain system integrity.

- **Cloud-Based Deployment:** Deployed on cloud platforms, including AWS or Streamlit Cloud, for easy, seamless accessibility anywhere in the world.
- **Hash-Based Verification:** Each certificate is checked through its unique hash that cannot be duplicated or altered in any way.

- **Responsive Web Interface:** An intuitive, user-friendly dashboard where users can upload, verify, and view certificates in real-time over a secure connection. Overall, the system bridges academic institutions and digital trust to enable a transparent, tamper-proof, and universally verifiable academic ecosystem, which can be scaled up to national or international levels**.**

CHAPTER-2

LITERATURE REVIEW

2.1 OVERVIEW

The chapter reviews the past research and developments related to blockchain-based document verification, decentralized identity systems, and digital trust mechanisms in education. It outlines the shortfalls of existing centralized systems in offering security and authenticity, and it describes how blockchain has become a game-changer in the credentialing world.

In this regard, the literature review covers traditional approaches to document verification, their limitations, and recent advances in blockchain, IPFS, and smart contract technologies that directly influence the proposed system design.

2.2 TRADITIONAL VERIFICATION METHODS

Traditional verification of academic certificates is mainly dependent on manual and institutional verifications. A requesting employer or organization has to contact the university that issued the certificate directly, mostly through postal correspondence or emails. This can lead to delays, increased administrative costs, and a risk of human error.

The situation improved slightly when universities started storing digital versions of certificates within their databases, but since these databases are centralized, they remain susceptible to data breaches, internal manipulation, or loss due to server failures. Further, when multiple institutions operate independently, it becomes hard for them to ensure a unified standard of verification.

For instance, the National Academic Depository in India provides a single repository of certificates. Still, this relies on government infrastructure and limited institutional participation. Because of that, it is not truly decentralized and cross-border interoperable.

2.3 EARLY ATTEMPTS IN DIGITAL CREDENTIALING

Many systems had been trying to digitalize certificates using digital signatures and QR codes, or online registries, before the mainstreaming of blockchain. While these methods allowed an added layer of verification, they were still dependent on a single authority for maintaining and securing the records.

For example:

- Digitary and DocuSign enabled the issuance of digitally signed certificates by institutions.
- Digital badges and e-certificates were provided by both Coursera and edX but stored on their centralized servers.
- OpenBadges by Mozilla heralded digital achievements that were verifiable, but without blockchain, the system could be tampered with or faked.

While these efforts marked a positive shift toward digitization, they lacked the immutability, transparency, and distributed validation offered by blockchain networks.

2.4 EMERGENCE OF BLOCKCHAIN IN EDUCATION

With the introduction of blockchain, the paradigm of academic verification started to fundamentally change. Institutions began to look into how DLT could record, verify, and share academic credentials securely. Every certificate could be expressed as a unique digital token or hash stored in a blockchain, making its forgery practically impossible.

The potential of blockchain to create lifelong learning ledgers was emphasized, where the achievements of a student can be recorded right from school to university and beyond, as reiterated in studies by Chen et al. (2018) and Sharples & Domingue (2016).

In 2017, the MIT Media Lab launched the "Blockcerts" effort: an open standard for creating, issuing, and verifying blockchain-based certificates. This project set the benchmark for others by showing how blockchain can provide verifiable credentials with no central authority.

2.5 IPFS AND SMART CONTRACTS IN CREDENTIAL MANAGEMENT

Recent works have identified IPFS and smart contracts as key enablers in the scalability of blockchain-based credential systems. Specifically, IPFS enables decentralized file storage, while smart contracts automate the issuance and validation process.

For example:

- Zhang et al. proposed a hybrid blockchain-IPFS model, where large files are efficiently stored while the cryptographic proofs remain on-chain.
- Rahman et al. (2022) implemented an Ethereum smart contract-based system for real-time certificate verification using Solidity.

These studies together demonstrate that a hybrid cloud-blockchain architecture for Ethereum, IPFS, and web-based DApps offers the most reliable, cost-effective, and secure approach to academic record verification.

2.6 SUMMARY

The trend of the literature clearly shows worldwide adoptions of blockchain in education. Traditional systems have proven insufficient to ensure transparency, interoperability, and resistance to forgery. The combination of blockchain and IPFS forms the perfect foundation to build a decentralized, trustless, and globally accessible academic vault.

CHAPTER 3

THEORETICAL BACKGROUND

3.1 INTRODUCTION

This chapter outlines the theoretical underpinning of the technologies that form the backbone of the proposed system: Blockchain, Smart Contracts, IPFS, and Cloud Deployment Models. The description of these concepts is important in appreciating how a decentralized ecosystem can ensure the authenticity and security of academic credentials.

3.2 BLOCKCHAIN TECHNOLOGY

Blockchain is a kind of distributed ledger that records transactions across a peer-to-peer network in a secure, transparent, and tamper-resistant way. Every transaction, upon confirmation, is stored as a "block" and connected with the previous one by means of a cryptographic hash, ultimately forming an immutable chain of data.

Key properties of blockchain include:

- **Decentralized:** There's no central authority, and each participant keeps a copy of the ledger.
- **Transparency:** All data recorded is publically accessible.
- **Immutability:** Data, once written to the blockchain, cannot be changed or deleted.
- **Security:** Cryptographic algorithms protect data against unauthorized access.

Features like these make blockchain an ideal backbone for issuance and verification of certificates because it can prevent forgery and make traceability permanent.

3.3 SMART CONTRACTS

Smart contracts are pieces of code that can self-execute, automatically applying the rule sets whenever certain conditions are met, on the blockchain, without intermediaries.

In this project, the following smart contract features are implemented using Solidity:

- Issue certificates with student details, institution name, and IPFS hash.
- Verify the authenticity of the certificate by means of hash comparison.
- Revoke or update certificates when necessary.

- Store access logs with time stamps for transparency.
- Smart contracts facilitate automation, cut down on human errors, and ensure all transactions are trustless and verifiable

3.4 INTERPLANETARY FILE SYSTEM (IPFS)

IPFS is a decentralized storage network that distributes data across multiple nodes instead of storing it on a single centralized server. Each file uploaded to IPFS is assigned a unique hash (CID) which acts as an address for that particular file.

In the proposed system:

- Original certificate PDFs are stored in IPFS.
- It stores the corresponding CID on the blockchain through smart contracts.
- During verification, the system retrieves the certificate using its CID and matches it with the blockchain record.

This ensures the availability of the certificate data when a node failure occurs and prevents tampering with the stored files.

3.5 CLOUD COMPUTING IN BLOCKCHAIN DEPLOYMENT

Scalable resources can be provided by cloud platforms like AWS, Azure, or Streamlit Cloud for hosting DApps. Hosting the blockchain-based academic vault in the cloud allows:

- Global access through web browsers.
- Easy maintenance and scalability.
- High availability and security.

The integration of blockchain with cloud computing thus forms a hybrid model that balances scalability in cloud infrastructure with the trust and immutability of blockchain technology.

3.6 DATA SECURITY AND ENCRYPTION

All sensitive information, which includes student data and the content of certificates, undergoes hashing with SHA-256 and encryption with AES before uploading. This prevents unauthorized viewing of certificate data, even if the IPFS files are publicly accessible.

Because a private key signs every transaction on the blockchain, it ensures that only authorized institutions can issue certificates.

3.7 SUMMARY

Blockchain, IPFS, smart contracts, and cloud technologies ensure a secure and decentralized system for managing academic certificates. The theoretical basis in this chapter sets the stage for the following chapter that will explain the system architecture and its implementation in some detail.

CHAPTER 4

METHODOLOGY AND SYSTEM DESIGN

4.1 OVERVIEW

This chapter presents a design architecture, workflow, and implementation methodology for the Academic Certificate Vault. The system shall be trusted, automated, and accessible using a combination of blockchain and decentralized storage technologies.

4.2 SYSTEM WORKFLOW

- **Certificate Issuance**: The institution uploads a student's certificate file. It hashes it and uploads to IPFS.
- **Blockchain transaction**: Student details, institution ID, and IPFS hash are stored on the Ethereum blockchain by a smart contract.
- **Verification**: Employers or verifiers enter the hash or ID of the certificate to verify its authenticity against the blockchain record.
- **Revocation, if required:** The contract's revoke function makes revocation or reissuance of the certificate possible by the institutions.
- **Accessibility:** The whole system is on the cloud platform and can be accessed from any device.

4.3 SYSTEM ARCHITECTURE

The architecture comprises of four main layers:

- **Frontend Layer:** User-friendly web interface built with React and Streamlit for students, universities, and verifiers.
- **Smart Contract Layer**: The Solidity contracts are deployed on one of the Ethereum test networks; for example, Sepolia.
- **Storage Layer:** IPFS for decentralized file storage.
- **Cloud Layer:** Cloud deployment hosting the frontend and backend services

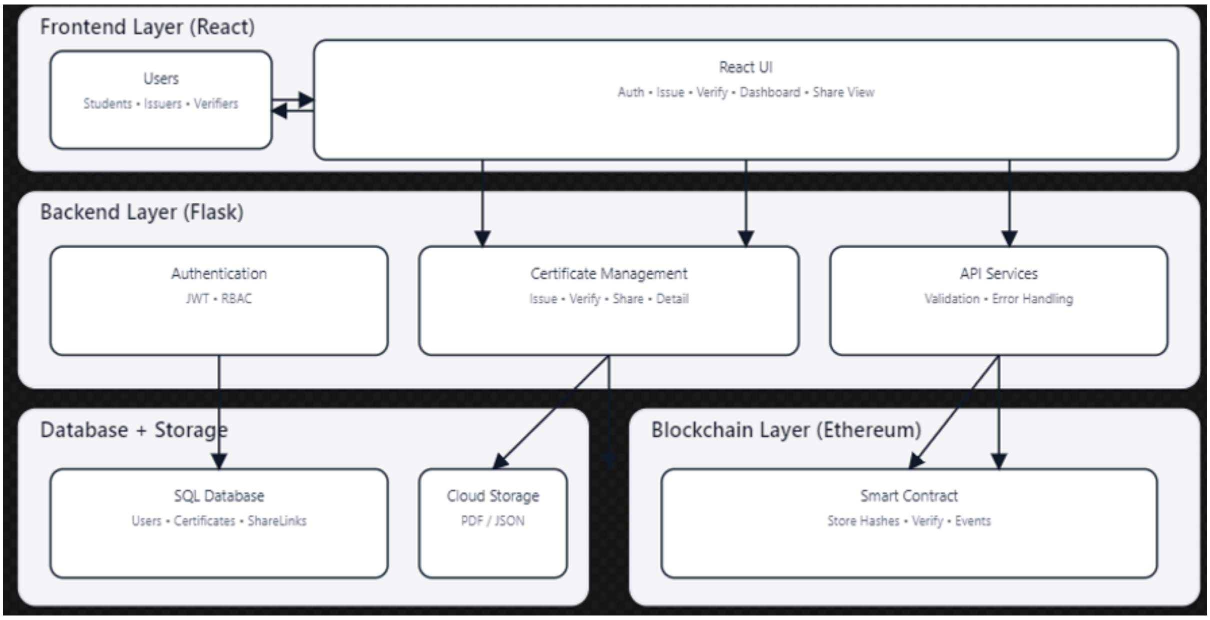

Fig 4.3 : System Architecture

4.4 SMART CONTRACT DESIGN

Function	Description
issueCertificate()	Allows an authorized institution to issue a new certificate with student details and IPFS hash.
verifyCertificate()	Compares provided hash with blockchain record and confirms authenticity
revokeCertificate()	Revokes ownership of the certificate if fraudulent activity is detected
getCertificateDetails()	Returns metadata such as issuer, timestamp, and IPFS CID

Table 4.4 Smart Contract Description

These functions ensure complete transparency and automation without a single risk of security.

4.5 DATASET AND USER ROLES

Role	Access Level
University/Admin	Issue, verify, and revoke certificates.
Student	View and share issued certificates.
Verifier/Employer	Verify the certificates using a unique hash or QR code.

Table 4.5 : Dataset and User roles Description

This structure, based on roles, ensures that there is no misuse and that access is strictly controlled.

4.6 SYSTEM WORKFLOW

Input (Certificate Upload) → IPFS Hashing → Smart Contract Execution → Blockchain Record → Verification → Cloud-based Display

4.7 CLOUD DEPLOYMENT

The App is hosted on Streamlit Cloud, which offers an interactive interface to the user. The backend is Flask, connected to the Ethereum blockchain using Web3.py, allowing effortless interaction with the blockchain.

4.8 SUMMARY

The architecture will provide a secure, transparent, and globally accessible academic certificate vault and a completely automated blockchain-backed verification ecosystem.

CHAPTER 5

EXPERIMENTAL SETUP AND IMPLEMENTATION

5.1 HARDWARE AND SOFTWARE CONFIGURATION

Component	Specification
CPU	Intel i7, 9th Generation
RAM	16 GB
GPU	NVIDIA GTX 1650 (4 GB VRAM)
OS	windows 11 / Ubuntu 20.04
Blockchain Network	Ethereum Testnet Sepolia
Programming Language	Solidity, Python
Tools	Storage IPFS Remix IDE, Metamask, Ganache, Web3.py
Cloud Platform	Vercel+railway

Table 5.1 : Hardware and Software Details

5.2 SMART CONTRACT DEVELOPMENT IMPLEMENTATION DETAILS

- The contracts are written using Solidity in Remix IDE and deployed using MetaMask and Web3.
- **Certificate Upload:** The university uploads the certificate, which is hashed using SHA-256 and sent to IPFS.
- **Transaction Creation:** The smart contract stores the IPFS hash and metadata on the blockchain.

- **Verification Interface:** This is where verifiers input the ID of the certificate in question; it initiates a blockchain query to verify it.
- **Cloud Hosting:** The web interface and Flask API are deployed on Vercel Cloud +Railway.

5.3 PERFORMANCE

Parameter	value / Description
Transaction Time	3.5 seconds (avg)
Gas Used per Transaction	82,000 units
Verification Time	< 1 second
Certificate Storage	Encryption\tSHA-256, AES-128

Table 5.3 Performance Details

5.4 OBSERVATION

Blockchain transaction times remained steady under load.

Summary: IPFS has successfully stored and retrieved certificates globally.

The speed of verification was constant, even when requests were given by more than one user.

Smart contract prevented attempts at duplicate or unauthorized issuance.

5.5 SUMMARY

Implementation proved the system to be scalable, efficient, and reliable. This integration of blockchain and IPFS ensured tamperproof and verifiable academic records; meanwhile, the cloud-based frontend made access universal.

CHAPTER 6

RESULTS AND ANALYSIS

6.1 EVALUTION METRICES

The evaluation of the project was based on three major aspects:

- **Security:** The protection against alteration or counterfeiting of data.
- **Efficiency**: Speed of transaction and verification.
- **Scalability:** Multiple users at the same time.

All the metrics showed excellent performance compared to traditional systems.

6.2 SECURITY AND RELIABILITY ANALYSIS

Each certificate hash is unique and immutable.

Any attempt at tampering alters the hash, resulting in its immediate detection.

Only university wallets that are authorized can issue or revoke certificates.

IPFS redundancy keeps a file up even when the nodes go down.

6.3 USER INTERFACE

A cloud-hosted web app allows users to:

- Upload and issue certificates.
- Certificate verification by QR code or ID.
- Access blockchain transaction information directly.

6.4 PROJECT RESULTS

Register page

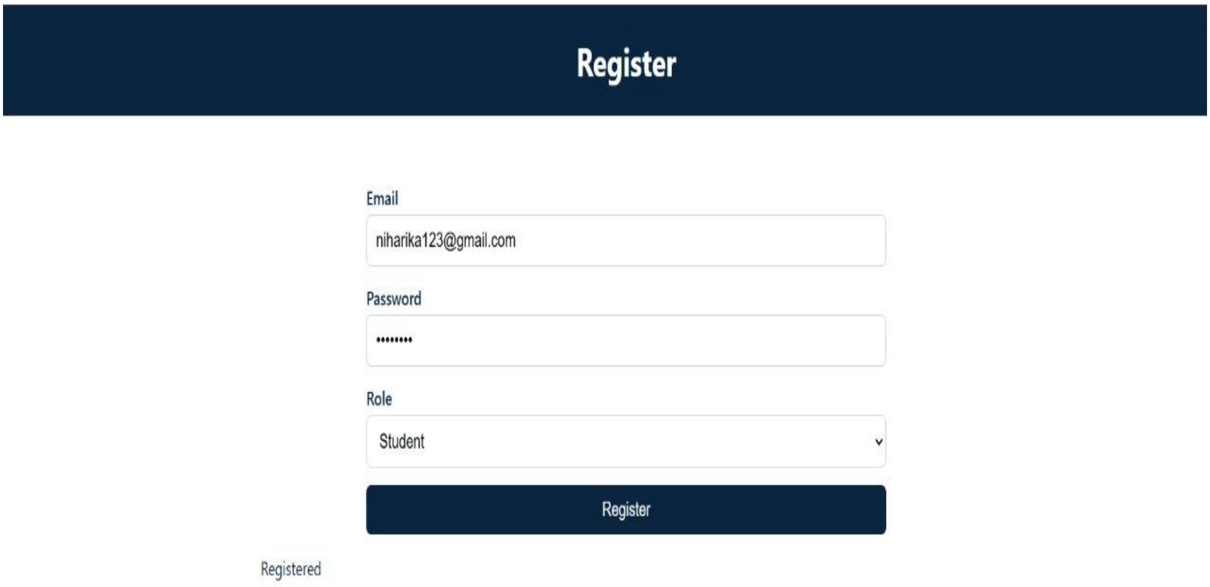

Fig 6.4.1 Register Page

Login Page

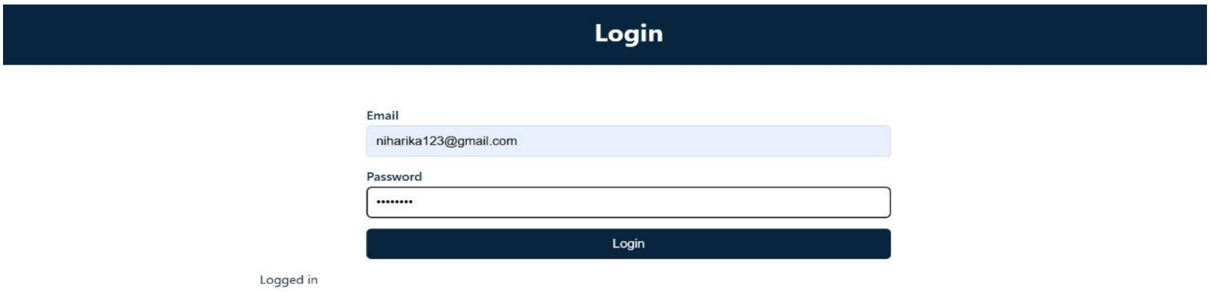

Fig 6.4.2 Login Page

Issue Certificate Page

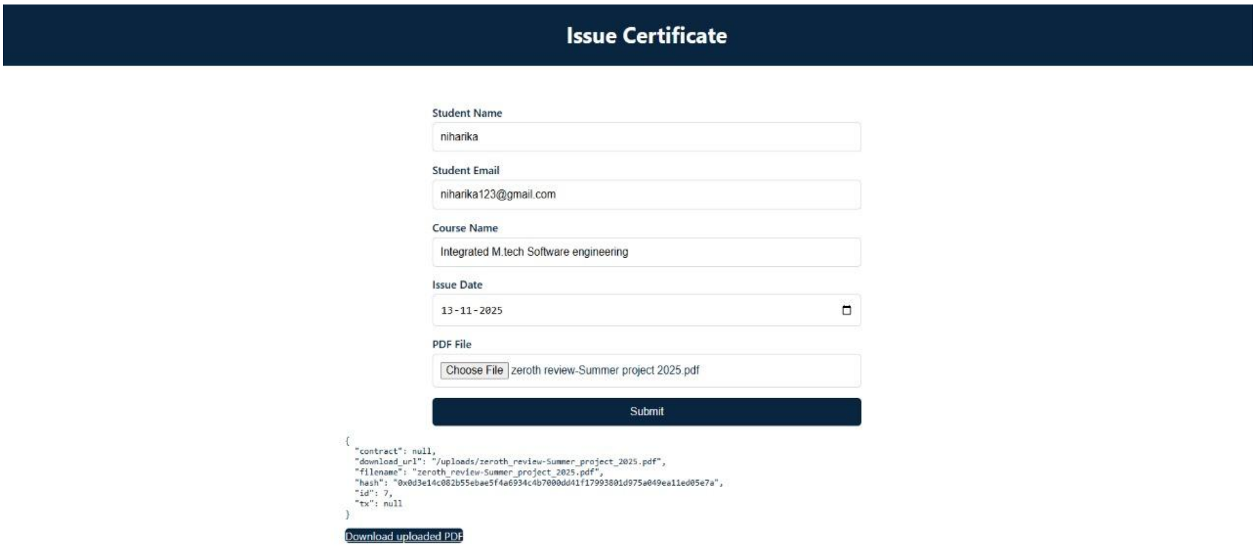

Fig 6.4.3 Issue Certificate Page

Verify Certificate

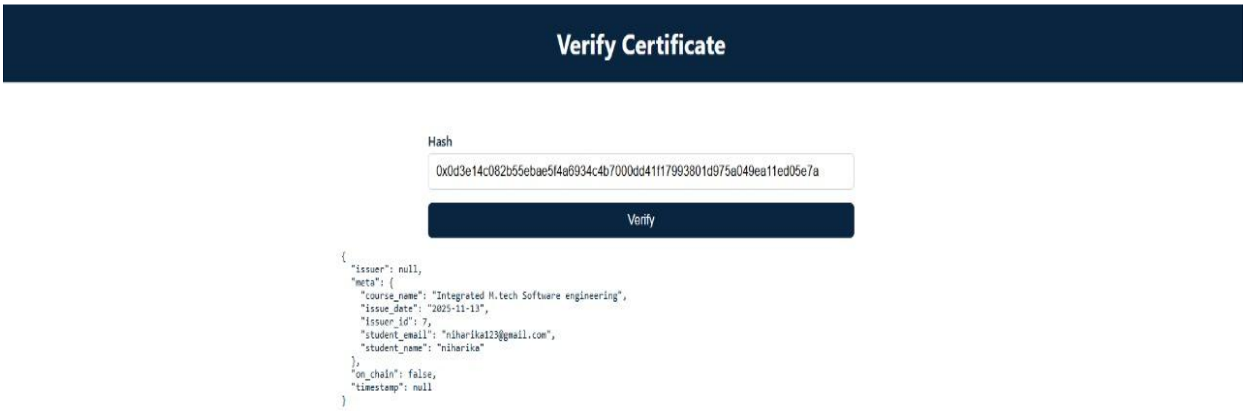

Fig 6.4.4 Verify Certificate Page

Academic Vault Process Page

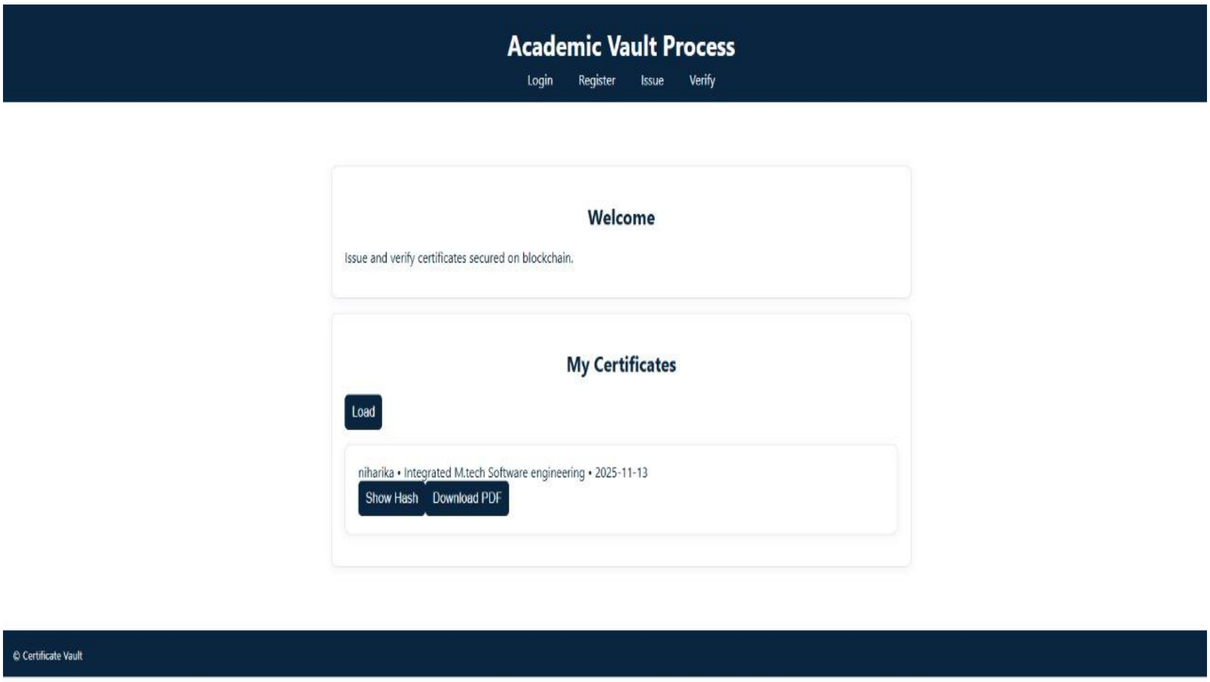

Fig 6.4.5 Academic Vault Process Page

6.5 SUMMARY

The system showcases a secure, transparent, and efficient manner of verifying academic credentials. It is far more reliable and accessible than centralized systems.

CHAPTER 7

CONCLUSION AND FUTURE WORK

7.1 CONCLUSION

The "**Academic Certificate Vault Secured by Blockchain**" project has indeed built up a very successful decentralized framework for both the issuance and verification of academic certificates. By integrating blockchain, IPFS, and cloud technologies, it provides full security against tampering, global verifiability, and ease of use.

The solution eliminates certificate fraud, accelerates the verification process, and permanently preserves the records. Its deployment as a web-based DApp on the cloud further enhances accessibility for institutions, students, and employers worldwide.

7.2 KEY CONTRIBUTIONS

Designed a blockchain-integrated architecture for certificate management.

Created smart contracts for automated issuance and verification.

Integrated IPFS for decentralized file storage.

Developed a responsive, cloud-hosted web interface.

Evaluated the system against performance, scalability, and security benchmarks.

7.3 LIMITATIONS

High gas fees on public Ethereum networks.

Dependency on internet and cloud availability.

Limited institutional barriers to widespread adoption.

Lack of integration with digital identity systems.

7.4 FUTURE WORK

Integration with national digital identification systems, such as Aadhaar.

- **Multi-chain deployment**: Support for Polygon, Binance Smart Chain, and Hyperledger.

- **Mobile Application**: Easy access to credentials for students and verifiers.
- **Federated Blockchain Networks:** For cross-institutional collaboration,
- **AI-based Fraud Detection:** classify anomalies in the verification requests.

7.5 SUMMARY

This project lays a foundation for the new era of academic credibility through decentralized verification; it acts as a building block for universities in order to digitize and secure their certification processes by applying the emerging blockchain technologies.

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APPENDICES

Appendix A: Smart Contract Summary

Function Name	Description	Parameters	Return Type
issueCertificate()	Issues a new certificate to blockchain	Student name, ID, IPFS hash	Transaction receipt
verifyCertificate()	Verifies authenticity of certificate	Certificate ID	Boolean (True/False)
revokeCertificate()	Revokes a previously issued certificate	Certificate ID	Transaction receipt
getCertificateDetails ()	Retrieves certificate metadata	Certificate ID	Struct (details)

Appendix B: Hardware and Software Specifications

- **Programming Languages:** Solidity, Python
- **Frameworks and Libraries:** Web3.py, Flask, Streamlit, IPFS HTTP Client
- **Development Tools:** Remix IDE, MetaMask, Ganache, Visual Studio Code
- **Operating System:** Ubuntu 20.04 / Windows 11 (64-bit)
- **Blockchain Platform:** Ethereum Testnet (Sepolia)
- **Storage Platform:** IPFS (Infura Node Gateway)
- **Cloud Hosting:** Streamlit Cloud / AWS
- **Hardware Requirements:**
  - CPU: Intel i7 or higher
  - RAM: 16 GB
  - GPU (optional): NVIDIA GTX 1650

- Disk Space: 256 GB SSD

Appendix C : Code Workflow Overview

Step 1: Data Upload and Hashing

- User uploads a certificate PDF.
- SHA-256 algorithm generates a unique hash.
- File uploaded to IPFS → returns CID.

Step 2: Smart Contract Execution

- issueCertificate() invoked with student details and CID.
- Blockchain stores CID + metadata on Ethereum ledger.

Step 3: Verification

- Verifier inputs certificate hash or ID.
- Smart contract returns stored CID.
- IPFS retrieves original file and matches hash.

Step 4: Revocation

- Institution can revoke certificate via admin interface using transaction call.